

MODUL PRAKTIK KEAMANAN SIBER

Modul 3 dari 3 — Defense in Depth Lab Project

Implementasi Wazuh SIEM

untuk Deteksi dan Respons Brute Force SSH

Centralized Log Monitoring dan Security Alert dengan Wazuh SIEM

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Environment	LAB-NET 192.168.10.0/24
Modul	Modul 3 — Wazuh SIEM Integration
Prasyarat	Modul 1 (Defense in Depth) + Modul 2 (Brute Force IR)

Kelanjutan dari:

Modul 1: Implementasi Defense in Depth (UFW + SSH Hardening + Fail2ban)

Modul 2: Brute Force Attack Simulation & Incident Response (NIST SP 800-61)

Modul ini menambahkan layer keempat Defense in Depth yaitu SIEM (Security Information and Event Management) menggunakan Wazuh sebagai platform monitoring terpusat.

1. Project Overview & Objectives

Wazuh SIEM sebagai Layer Keempat Defense in Depth

Latar Belakang

Pada Modul 1, kamu telah membangun sistem keamanan berlapis (Defense in Depth) menggunakan UFW Firewall, SSH Hardening, dan Fail2ban. Pada Modul 2, kamu telah membuktikan sistem tersebut mampu mendeteksi dan menahan serangan brute force SSH secara otomatis melalui simulasi menggunakan Hydra.

Namun, pertanyaannya adalah: bagaimana SOC analyst bisa memantau semua kejadian itu dari satu dashboard terpusat? Inilah fungsi SIEM (Security Information and Event Management).

Modul 3 ini mengintegrasikan Wazuh sebagai SIEM engine yang akan mengumpulkan, menganalisis, dan menampilkan semua security event dari Ubuntu Server secara real-time melalui dashboard berbasis web.

Objectives

1. Menginstal dan mengkonfigurasi Wazuh SIEM dalam mode single-node (all-in-one).
2. Menghubungkan Ubuntu Server sebagai agent yang dimonitor Wazuh.
3. Memverifikasi bahwa Wazuh mampu mendeteksi percobaan login SSH yang gagal secara real-time.
4. Menjalankan ulang simulasi brute force dan mengamati alert di Wazuh Dashboard.
5. Mengkorelasikan event Fail2ban dengan alert yang muncul di SIEM.
6. Mendokumentasikan incident response berbasis SIEM sesuai perspektif SOC analyst.

Lab Environment

Host	IP Address	Role	OS
Ubuntu Server	192.168.10.100	Defender + Wazuh SIEM	Ubuntu Server 24.04 LTS
Ubuntu Desktop	192.168.10.101	Attacker (Hydra)	Ubuntu Desktop 24.04

Arsitektur Defense in Depth — Layer Overview

Layer	Komponen	Fungsi
1	UFW Firewall	Membatasi port — hanya 22 dan 80 yang terbuka (Modul 1)
2	SSH Hardening	PermitRootLogin no, AllowUsers ubuntu (Modul 1)
3	Fail2ban	Auto-block IP setelah maxretry gagal login (Modul 1 & 2)
4 (BARU)	Wazuh SIEM	Centralized logging, real-time alert, SOC dashboard (Modul 3)

2. Phase 1 — Persiapan Sistem

Verifikasi baseline sebelum instalasi Wazuh

Sebelum menginstal Wazuh, pastikan semua layanan yang sudah ada dari Modul 1 dan 2 masih berjalan dengan baik. Seluruh perintah di section ini dijalankan di Ubuntu Server (192.168.10.100).

2.1 Update dan Upgrade Sistem

Pastikan Ubuntu Server dalam kondisi terbaru sebelum menginstal Wazuh. Koneksi internet diperlukan (via adapter NAT di VirtualBox).

```
# Jalankan di: Ubuntu Server (192.168.10.100)
sudo apt update && sudo apt upgrade -y

# Verifikasi koneksi internet aktif:
ping google.com -c 3
```

⚠ Pastikan VirtualBox Adapter 2 di Ubuntu Server dikonfigurasi sebagai NAT untuk akses internet. Adapter 1 tetap Internal Network (LAB-NET) untuk komunikasi antar VM.

```
Users logged in:      0
IPv4 address for enp0s8: 10.0.3.15
IPv6 address for enp0s8: fd17:625c:f037:3:a00:27ff:fea1:c8aa

Expanded Security Maintenance for Applications is not enabled.

147 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ubuntu@UbuntuServer:~$ sudo apt update && apt upgrade -y
[sudo] password for ubuntu:
Hit:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,472 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [237 kB]
42% [6 Translation-en store 0 B]
```

```
ubuntu@UbuntuServer:~$ ping google.com -c 3
PING google.com (74.125.24.102) 56(84) bytes of data:
64 bytes from sf-in-f102.1e100.net (74.125.24.102): icmp_seq=1 ttl=255 time=57.8 ms
64 bytes from sf-in-f102.1e100.net (74.125.24.102): icmp_seq=2 ttl=255 time=49.0 ms
64 bytes from sf-in-f102.1e100.net (74.125.24.102): icmp_seq=3 ttl=255 time=42.2 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 42.217/49.669/57.777/6.369 ms
ubuntu@UbuntuServer:~$ _
```

2.2 Verifikasi Status Layanan Baseline

Cek semua layanan yang sudah dikonfigurasi dari modul sebelumnya. Semua harus berstatus active (running).

```
# Jalankan di: Ubuntu Server
# Step 1 — Cek SSH
sudo systemctl status ssh
```

```
ubuntu@UbuntuServer:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab
   Active: active (running) since Thu 2026-02-19 07:19:18 UTC; 3min 26s ago
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 790 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 826 (sshd)
     Tasks: 1 (limit: 4606)
    Memory: 2.1M (peak: 2.4M)
       CPU: 56ms
   CGroup: /system.slice/ssh.service
           └─826 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 19 07:19:17 UbuntuServer systemd[1]: Starting ssh.service - OpenBSD Secure
Feb 19 07:19:17 UbuntuServer sshd[826]: Server listening on 0.0.0.0 port 22.
Feb 19 07:19:18 UbuntuServer sshd[826]: Server listening on :: port 22.
Feb 19 07:19:18 UbuntuServer systemd[1]: Started ssh.service - OpenBSD Secure S
lines 1-18/18 (END)
```

```
# Step 2 —Cek Fail2ban
sudo systemctl status fail2ban
```

```
ubuntu@UbuntuServer:~$ sudo systemctl status fail2ban
• fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset:
   Active: active (running) since Thu 2026-02-19 07:19:16 UTC; 4min 6s ago
   Docs: man:fail2ban(1)
   Main PID: 774 (fail2ban-server)
     Tasks: 5 (limit: 4606)
    Memory: 42.6M (peak: 42.8M)
       CPU: 1.243s
   CGroup: /system.slice/fail2ban.service
           └─774 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 19 07:19:16 UbuntuServer systemd[1]: Started fail2ban.service - Fail2Ban Se
Feb 19 07:19:17 UbuntuServer fail2ban-server[774]: 2026-02-19 07:19:17,593 fail
Feb 19 07:19:19 UbuntuServer fail2ban-server[774]: Server ready
lines 1-14/14 (END)
```

```
# Step 3 — Cek UFW
sudo ufw status verbose
```

```
ubuntu@UbuntuServer:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp (OpenSSH) ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
22/tcp (OpenSSH (v6)) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)

ubuntu@UbuntuServer:~$
```

```
# Step 4 — Cek Jail SSH (pastikan 0 banned sebelum mulai)
sudo fail2ban-client status sshd
```

```
ubuntu@UbuntuServer:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed: 0
|   \- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
\-- Actions
    |- Currently banned: 0
    |- Total banned: 0
    \- Banned IP list:
ubuntu@UbuntuServer:~$
```

2.3 Konfigurasi Adapter VirtualBox untuk Internet

Wazuh diinstal dari internet. Kamu perlu menambahkan Adapter 2 sebagai NAT di Ubuntu Server VM sebelum memulai instalasi.

Langkah di VirtualBox:

1. Matikan Ubuntu Server VM terlebih dahulu (sudo poweroff).
2. Di VirtualBox, pilih Ubuntu Server VM, klik Settings.
3. Masuk ke tab Network, klik Adapter 2.
4. Centang Enable Network Adapter, pilih Attached to: NAT.
5. Klik OK, lalu nyalakan Ubuntu Server VM kembali.

```
# Setelah VM nyala, verifikasi adapter baru terdeteksi:
ip a
```

```
# Kamu akan melihat interface baru (misal enp0s8) dengan IP dari NAT
# Biasanya: 10.0.x.x/24
```

```
valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
group default qlen 1000
    link/ether 08:00:27:a1:c8:aa brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 metric 100 brd 10.0.3.255 scope global dynamic enp0s8
        valid_lft 85871sec preferred_lft 85871sec
    inet6 fd17:625c:f037:3:a00:27ff:fea1:c8aa/64 scope global dynamic mngtmpaddr
        noprefixroute
        valid_lft 86303sec preferred_lft 14303sec
    inet6 fe80::a00:27ff:fea1:c8aa/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@UbuntuServer:~$
```

```
# Tes koneksi internet:
ping 8.8.8.8 -c 3
```

```
ubuntu@UbuntuServer:~$ ping 8.8.8.8 -c 3
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=68.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=45.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=39.6 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 39.566/50.960/68.022/12.288 ms
ubuntu@UbuntuServer:~$
```

3. Phase 2 — Instalasi Wazuh SIEM

Single Node Deployment (All-in-One)

Apa itu Wazuh?

Wazuh adalah platform SIEM open-source yang terdiri dari tiga komponen utama:

Wazuh Manager: Menerima dan memproses log dari agent, menjalankan rules/deteksi

Wazuh Indexer: Database berbasis OpenSearch untuk menyimpan security events

Wazuh Dashboard: Interface web berbasis OpenSearch Dashboards untuk visualisasi alert

Mode All-in-One berarti ketiga komponen di atas diinstal di satu server (192.168.10.100).

Ubuntu Server akan berperan sebagai SIEM sekaligus host yang dimonitor.

3.1 Download Wazuh Installer

```
# Jalankan di: Ubuntu Server
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
```

```
# Verifikasi file berhasil didownload:
ls -lh wazuh-install.sh
```

```
ubuntu@UbuntuServer:~$ ls -lh
total 168K
drwxrwxr-x 2 ubuntu ubuntu 4.0K Feb 18 08:43 latian_bash
-rw-rw-r-- 1 ubuntu ubuntu 162K Feb 19 07:30 wazuh-install.sh
ubuntu@UbuntuServer:~$ _
```

3.2 Instalasi Wazuh All-in-One

⚠ Proses instalasi ini membutuhkan waktu 10-30 menit tergantung kecepatan internet dan spesifikasi VM. Jangan tutup terminal atau matikan VM selama proses berlangsung.

```
# Jalankan di: Ubuntu Server
sudo bash wazuh-install.sh -a --ignore-check
```

```
ubuntu@UbuntuServer:~$ sudo bash wazuh-install.sh -a --ignore-check
19/02/2026 07:32:41 INFO: Starting Wazuh installation assistant. Wazuh version:
4.7.5
19/02/2026 07:32:41 INFO: Verbose logging redirected to /var/log/wazuh-install.1
og
```

```
# Di akhir instalasi, kamu akan mendapatkan output seperti:
# 07/08/2026 10:30:00 INFO: --- Summary ---
# 07/08/2026 10:30:00 INFO: You can access the web interface https://192.168.10.100
# 07/08/2026 10:30:00 INFO: User: admin
# 07/08/2026 10:30:00 INFO: Password: <auto-generated-password>
# 07/08/2026 10:30:00 INFO: Installation finished.

# PENTING: Catat username dan password yang muncul!
```

💡 Catat password Wazuh yang muncul di akhir instalasi. Password ini auto-generated dan tidak bisa dilihat lagi tanpa reset. Kamu bisa juga menemukannya di file: `sudo tar -O -xvf wazuh-install-files.tar wazuh-passwords.txt`

```
19/02/2026 08:04:08 INFO: Initializing Wazuh dashboard web application.
19/02/2026 08:04:08 INFO: Wazuh dashboard web application not yet initialized. Waiting...
19/02/2026 08:04:24 INFO: Wazuh dashboard web application not yet initialized. Waiting...
19/02/2026 08:04:39 INFO: Wazuh dashboard web application initialized.
19/02/2026 08:04:39 INFO: --- Summary ---
19/02/2026 08:04:39 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: 33V2m.Yts5fyJ5XF*vv0VcHktlWjtPWg
19/02/2026 08:04:39 INFO: Installation finished.
ubuntu@UbuntuServer:~$
```

4. Phase 3 — Verifikasi Service dan Dashboard

Memastikan semua komponen Wazuh aktif

4.1 Verifikasi Status Tiga Service Utama Wazuh

Wazuh terdiri dari tiga service yang harus semuanya berstatus active (running).

```
# Jalankan di: Ubuntu Server
sudo systemctl status wazuh-manager
```

```
ubuntu@UbuntuServer:~$ systemctl status wazuh-manager
• wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; pro
   Active: active (running) since Thu 2026-02-19 07:55:46 UTC; 16min ago
     Tasks: 111 (limit: 4606)
    Memory: 257.8M (peak: 476.0M)
       CPU: 57.655s
    CGroup: /system.slice/wazuh-manager.service
            └─47554 /var/ossec/framework/python/bin/python3 /var/ossec/api/scrip
              └─47594 /var/ossec/bin/wazuh-authd
                └─47609 /var/ossec/bin/wazuh-db
                  └─47619 /var/ossec/bin/wazuh-execd
                    └─47633 /var/ossec/bin/wazuh-analysisd
                      └─47657 /var/ossec/bin/wazuh-syscheckd
                        └─47673 /var/ossec/bin/wazuh-remoted
                          └─47724 /var/ossec/bin/wazuh-logcollector
                            └─47740 /var/ossec/bin/wazuh-monitord
                              └─47749 /var/ossec/bin/wazuh-modulesd
                                └─47781 /var/ossec/framework/python/bin/python3 /var/ossec/api/scrip
                                  └─47784 /var/ossec/framework/python/bin/python3 /var/ossec/api/scrip
                                    └─47787 /var/ossec/framework/python/bin/python3 /var/ossec/api/scrip
```

```
sudo systemctl status wazuh-dashboard
```

```
ubuntu@UbuntuServer:~$ systemctl status wazuh-dashboard
• wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; pres
   Active: active (running) since Thu 2026-02-19 08:08:45 UTC; 9min ago
   Main PID: 49529 (node)
   Tasks: 11 (limit: 4606)
   Memory: 203.0M (peak: 267.4M)
   CPU: 20.811s
   CGroup: /system.slice/wazuh-dashboard.service
           └─49529 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --m
lines 1-9/9 (END)
```

```
sudo systemctl status wazuh-indexer
```

```
ubuntu@UbuntuServer:~$ systemctl status wazuh-indexer
• wazuh-indexer.service - Wazuh-indexer
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; pr
   Active: active (running) since Thu 2026-02-19 07:48:59 UTC; 24min ago
   Docs: https://documentation.wazuh.com
   Main PID: 3975 (java)
   Tasks: 65 (limit: 4606)
   Memory: 2.2G (peak: 2.2G)
   CPU: 1min 53.039s
   CGroup: /system.slice/wazuh-indexer.service
           └─3975 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopense
lines 1-10/10 (END)
```

4.2 Pastikan Service Enable (Auto-start)

```
# Pastikan semua service start otomatis saat reboot:
```

```
sudo systemctl enable wazuh-manager
sudo systemctl enable wazuh-dashboard
sudo systemctl enable wazuh-indexer
```

```
ubuntu@UbuntuServer:~$ systemctl enable wazuh-manager
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: ubuntu
Password:
==== AUTHENTICATION COMPLETE ====
ubuntu@UbuntuServer:~$ systemctl enable wazuh-dashboard
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: ubuntu
Password:
==== AUTHENTICATION COMPLETE ====
ubuntu@UbuntuServer:~$ systemctl enable wazuh-indexer
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: ubuntu
Password:
==== AUTHENTICATION COMPLETE ====
ubuntu@UbuntuServer:~$
```

PASTIKAN JUGA :

Jika sudah running, pastikan port 443 dibuka di UFW:

```
bash
sudo ufw allow 443/tcp
sudo ufw reload
sudo ufw status verbose
```

Verifikasi port 443 benar-benar listening:

```
bash
sudo ss -tlnp | grep 443
```

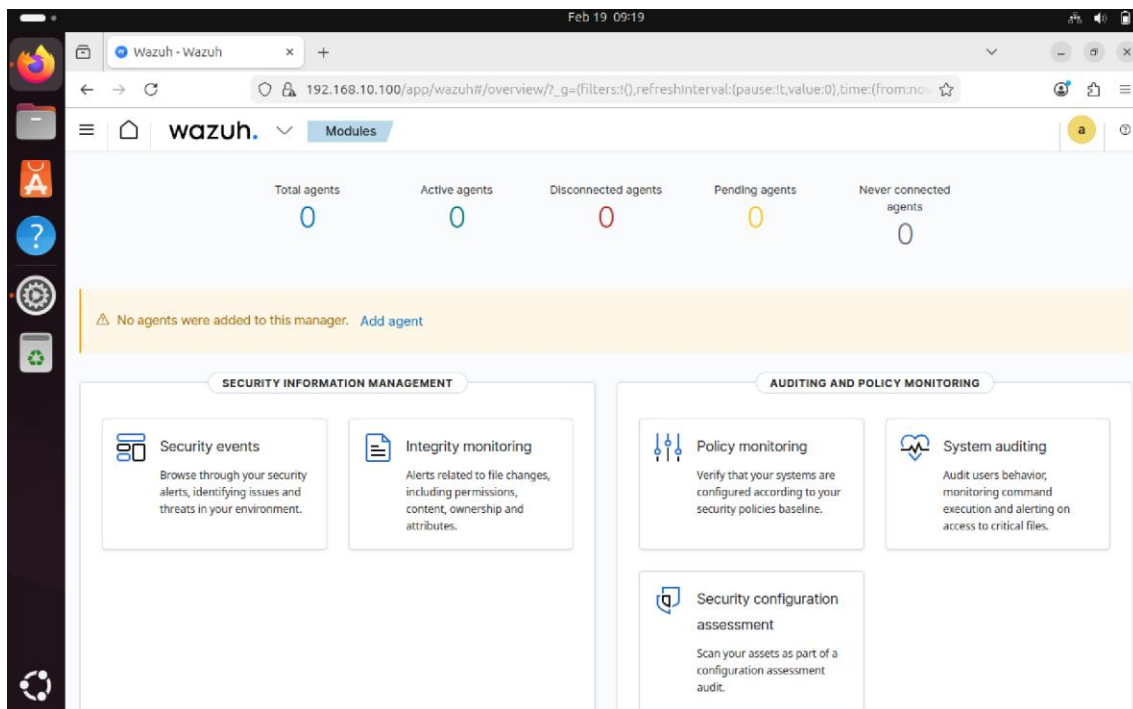
Ini dilakukan karena dimodul sebelumnya, port 443 diblock atau tidak dibuka.

4.3 Akses Wazuh Dashboard dari Ubuntu Desktop

Wazuh Dashboard diakses melalui browser dari Ubuntu Desktop (192.168.10.101). Karena menggunakan self-signed certificate, browser akan menampilkan peringatan keamanan — klik Advanced dan pilih Accept Risk untuk melanjutkan.

```
# Buka di: Ubuntu Desktop (192.168.10.101)
# Buka browser (Firefox/Chrome) dan ketik di address bar:
https://192.168.10.100

# Login dengan:
# Username: admin
# Password: <password dari output instalasi>
```



5. Phase 4 — Konfigurasi Monitoring Log SSH

Memastikan Wazuh memantau auth.log

5.1 Cek Konfigurasi ossec.conf

Wazuh Manager menggunakan file konfigurasi utama di `/var/ossec/etc/ossec.conf`. Kita perlu memastikan bahwa file `/var/log/auth.log` sudah terdaftar sebagai sumber log yang dipantau.

```
# Jalankan di: Ubuntu Server
sudo nano /var/ossec/etc/ossec.conf

# Cari bagian <localfile> yang memonitor auth.log
# Harus ada konfigurasi seperti ini:
#   <localfile>
#     <log_format>syslog</log_format>
#     <location>/var/log/auth.log</location>
#   </localfile>

# Keluar dari nano: CTRL+X
```

💡 Jika konfigurasi `/var/log/auth.log` belum ada, tambahkan secara manual di dalam blok `<ossec_config>`. Kemudian restart wazuh-manager agar perubahan aktif.

```
# Jika perlu menambahkan monitoring auth.log secara manual,
# tambahkan blok ini sebelum tag </ossec_config> penutup:

# <localfile>
#   <log_format>syslog</log_format>
#   <location>/var/log/auth.log</location>
# </localfile>

# Simpan file (CTRL+O, Enter, CTRL+X)

# Restart Wazuh Manager:
sudo systemctl restart wazuh-manager

# Verifikasi restart berhasil:
sudo systemctl status wazuh-manager
```

```
</ossec_config>

<ossec_config>
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/auth.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/syslog</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/kern.log</location>
  </localfile>
</ossec_config>
```

5.2 Verifikasi Wazuh Menerima Log dari Server

Cek apakah Wazuh Manager sudah mulai membaca auth.log dengan melihat log Wazuh sendiri.

```
# Jalankan di: Ubuntu Server
sudo tail -f /var/ossec/logs/ossec.log

# Kamu akan melihat log aktivitas Wazuh Manager
# Tekan CTRL+C untuk berhenti

# Cek apakah ada event yang sudah masuk:
sudo grep 'auth.log' /var/ossec/logs/ossec.log | head -10
```

```
ubuntu@UbuntuServer:~$ sudo tail -f /var/log/local/ lock/ log/
ubuntu@UbuntuServer:~$ sudo tail -f /var/ossec/logs/ossec.log
2026/02/19 09:01:41 wazuh-modulesd:control: INFO: Starting control thread.
2026/02/19 09:01:41 wazuh-modulesd:task-manager: INFO: (8200): Module Task Manager started.
2026/02/19 09:01:41 sca: INFO: Skipping policy '/var/ossec/ruleset/sca/cis_ubuntu22-04.yml': 'Check Ubuntu version.'
2026/02/19 09:01:41 sca: INFO: Security Configuration Assessment scan finished. Duration: 0 seconds.
2026/02/19 09:01:41 wazuh-modulesd:syscollector: INFO: Module started.
2026/02/19 09:01:41 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/02/19 09:01:42 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/02/19 09:01:47 wazuh-syscheckd: INFO: (6009): File integrity monitoring scan ended.
2026/02/19 09:01:47 wazuh-syscheckd: INFO: FIM sync module started.
2026/02/19 09:02:33 rootcheck: INFO: Ending rootcheck scan.
```

6. Phase 5 — Simulasi Brute Force SSH

Membuktikan deteksi Wazuh secara real-time

Sebelum Memulai Simulasi

Pastikan kondisi berikut terpenuhi:

1. Wazuh Dashboard terbuka di browser Ubuntu Desktop
2. Terminal Ubuntu Server memonitor: `sudo tail -f /var/ossec/logs/alerts/alerts.log`
3. Hydra tersedia di Ubuntu Desktop (dari Modul 2)
4. Fail2ban dalam kondisi aktif dan 0 banned IPs

6.1 Buka Monitoring Alert di Ubuntu Server

Sebelum menjalankan serangan, buka terminal terpisah di Ubuntu Server untuk memonitor alert Wazuh secara real-time.

```
# Jalankan di: Ubuntu Server — Terminal TERPISAH (Terminal B)
sudo tail -f /var/ossec/logs/alerts/alerts.log

# Biarkan terminal ini terbuka selama simulasi!
# Kamu akan melihat alert brute force muncul real-time di sini.
```

6.2 Reset Fail2ban Sebelum Simulasi

```
# Jalankan di: Ubuntu Server (Terminal A)
# Pastikan tidak ada IP yang masih terblokir dari sesi sebelumnya:
sudo fail2ban-client status sshd

# Jika ada IP yang masih terban, unban terlebih dahulu:
sudo fail2ban-client set sshd unbanip 192.168.10.101

# Restart fail2ban untuk fresh start:
sudo systemctl restart fail2ban

# Verifikasi:
```

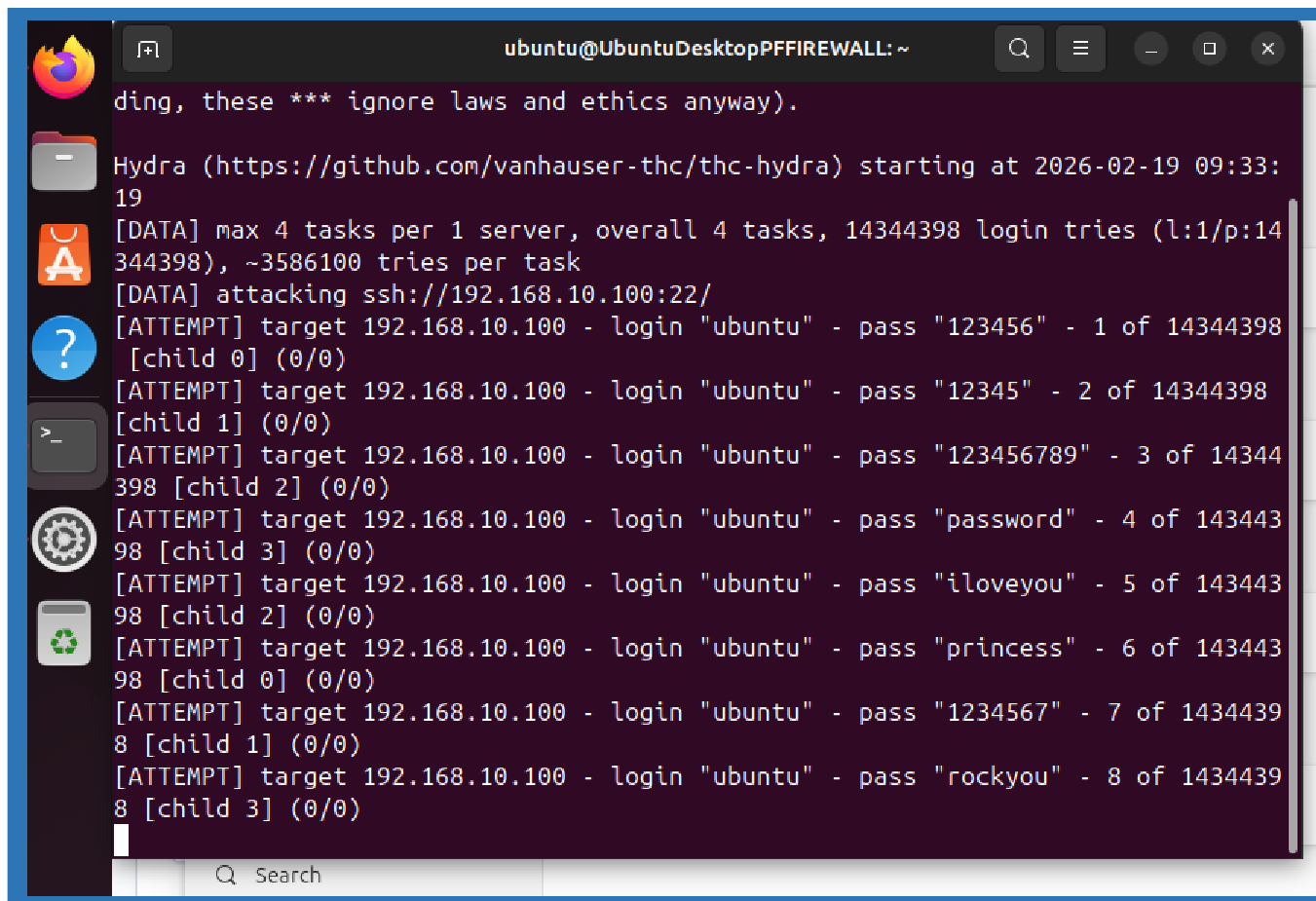
```
sudo fail2ban-client status sshd
# Expected: Currently banned: 0
```

```
ubuntu@UbuntuServer:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     0
|   \- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
|- Actions
|   |- Currently banned: 0
|   |- Total banned:     0
|   \- Banned IP list:
ubuntu@UbuntuServer:~$ _
```

6.3 Jalankan Serangan Brute Force dari Ubuntu Desktop

```
# Jalankan di: Ubuntu Desktop (Attacker — 192.168.10.101)
hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt \
      192.168.10.100 ssh -t 4 -V -f

# Penjelasan flag:
# -l ubuntu      : target username
# -P rockyou.txt : wordlist password
# -t 4           : 4 thread paralel
# -V            : verbose — tampilkan setiap percobaan
# -f            : stop setelah berhasil (tidak akan berhasil karena Fail2ban)
```



The screenshot shows a terminal window titled 'ubuntu@UbuntuDesktopPFFIREWALL: ~'. The output of the Hydra command is as follows:

```
ding, these *** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-19 09:33:19
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100 tries per task
[DATA] attacking ssh://192.168.10.100:22/
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "123456" - 1 of 14344398 [child 0] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "12345" - 2 of 14344398 [child 1] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "123456789" - 3 of 14344398 [child 2] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "password" - 4 of 14344398 [child 3] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "iloveyou" - 5 of 14344398 [child 2] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "princess" - 6 of 14344398 [child 0] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "1234567" - 7 of 14344398 [child 1] (0/0)
[ATTEMPT] target 192.168.10.100 - login "ubuntu" - pass "rockyou" - 8 of 14344398 [child 3] (0/0)
```

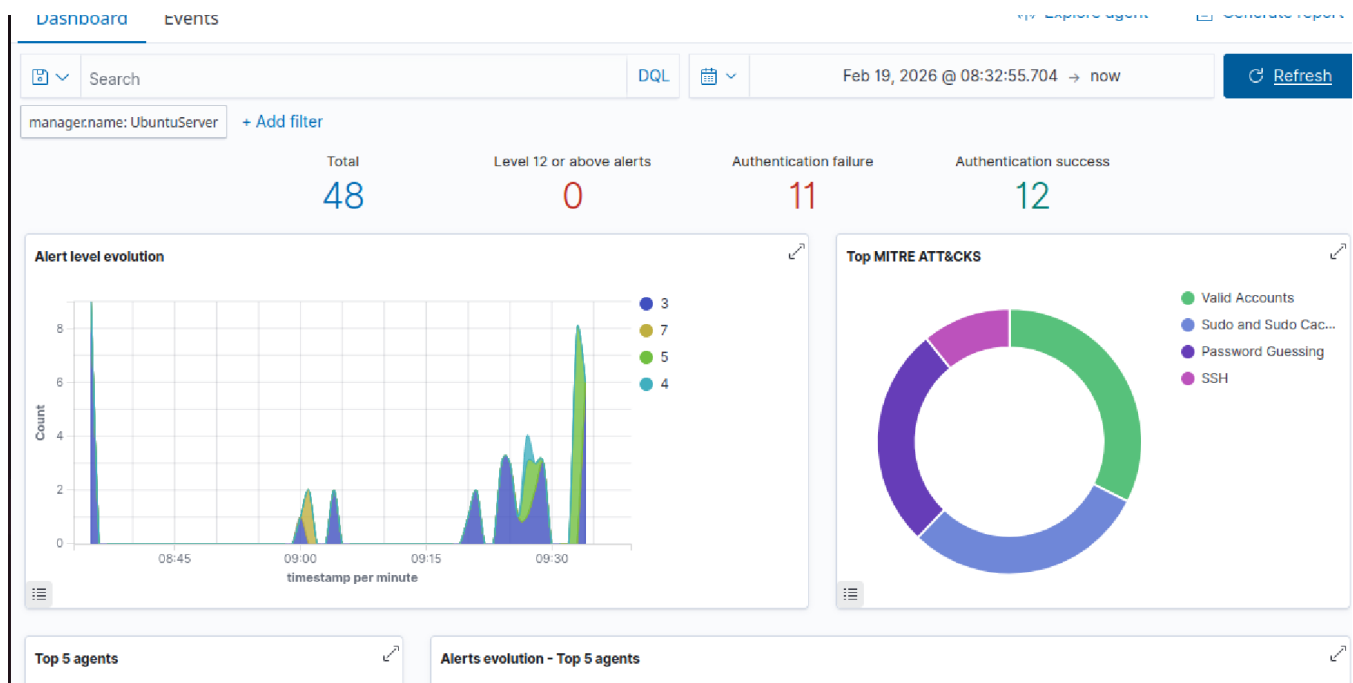
7. Phase 6 — Observasi Deteksi di Wazuh Dashboard

Melihat alert brute force di SIEM

7.1 Navigasi ke Security Events di Dashboard

Masuk ke browser di Ubuntu Desktop yang sudah terbuka di Wazuh Dashboard, lalu navigasi ke Security Events.

1. Di menu kiri Wazuh Dashboard, klik Security Events atau Threat Intelligence.
2. Pilih tab Authentication atau SSH untuk memfilter event login.
3. Atur time range ke 15 minutes last untuk melihat event terbaru.
4. Filter berdasarkan IP source: 192.168.10.101 (IP attacker Desktop).



7.2 Observasi Alert Brute Force

Wazuh memiliki rules bawaan untuk mendeteksi pola brute force SSH. Berikut rule IDs yang relevan:

Rule ID	Nama Rule	Deskripsi
5710	sshd: Attempt to login using a non-existent user	Percobaan login dengan username yang tidak ada
5712	sshd: POSSIBLE BREAK-IN ATTEMPT!	Wazuh mendeteksi pola mencurigakan
5720	Multiple SSH authentication failures	Banyak kegagalan autentikasi berturut-turut
5763	sshd: brute force trying to get access	Deteksi pola brute force otomatis
30101	fail2ban: banning host	Fail2ban memblokir IP (dari fail2ban.log)

Feb 19, 2026 @ 09:33:24.629	000	UbuntuServer	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Table JSON Rule							
@timestamp	2026-02-19T09:33:24.629Z						
_id	SP0_dZwBUQY9Nj9AUIPX						
agent.id	000						
agent.name	UbuntuServer						
data.dstuser	ubuntu						
data.srcip	192.168.10.101						
data.srcport	36776						
decoder.name	sshd						
decoder.parent	sshd						
full_log	2026-02-19T09:33:23.681784+00:00 UbuntuServer sshd[2614]: Failed password for ubuntu from 192.168.10.101 port 36776 ssh2						
_id	1771493604.34400						

7.3 Korelasi Event di Terminal alerts.log

Kembali ke Terminal B di Ubuntu Server yang masih menjalankan tail -f alerts.log. Kamu akan melihat alert yang sama dengan yang tampil di dashboard.

```

..AC.7,nist_800_53.AC.6,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:14 UbuntuServer->/var/log/auth.log
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
2026-02-19T09:34:14.702083+00:00 UbuntuServer sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/fail2ban-client status sshd
tty: ttty1
pwd: /home/ubuntu
command: /usr/bin/fail2ban-client status sshd

** Alert 1771493654.35973: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:14 UbuntuServer->/var/log/auth.log
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root(uid=0)
2026-02-19T09:34:14.708465+00:00 UbuntuServer sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
uid: 1000

** Alert 1771493656.36433: - pam,syslog,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:16 UbuntuServer->/var/log/auth.log
Rule: 5502 (level 3) -> 'PAM: Login session closed.'
User: root
2026-02-19T09:34:14.775989+00:00 UbuntuServer sudo: pam_unix(sudo:session): session closed for user root

** Alert 1771493662.36826: - syslog,sudo,pci_dss_10.2.5,pci_dss_10.2.2,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:22 UbuntuServer->/var/log/auth.log
Rule: 5402 (level 3) -> 'Successful sudo to ROOT executed.'
User: root
2026-02-19T09:34:22.755567+00:00 UbuntuServer sudo: ubuntu : TTY=ttty1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/fail2ban-client status sshd
tty: ttty1
pwd: /home/ubuntu
command: /usr/bin/fail2ban-client status sshd

** Alert 1771493662.37389: - pam,syslog,authentication_success,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:22 UbuntuServer->/var/log/auth.log
Rule: 5501 (level 3) -> 'PAM: Login session opened.'
User: root(uid=0)
2026-02-19T09:34:22.757349+00:00 UbuntuServer sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
uid: 1000

** Alert 1771493664.37849: - pam,syslog,pci_dss_10.2.5,gpg13_7.8,gpg13_7.9,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AU.14,nist_800_53_AC.7,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Feb 19 09:34:24 UbuntuServer->/var/log/auth.log
Rule: 5502 (level 3) -> 'PAM: Login session closed.'
User: root
2026-02-19T09:34:22.822631+00:00 UbuntuServer sudo: pam_unix(sudo:session): session closed for user root

```

a

8. Phase 7 — Korelasi Fail2ban dengan Wazuh

Verifikasi auto-block dan SIEM event

8.1 Konfirmasi Fail2ban Berhasil Ban IP

```

# Jalankan di: Ubuntu Server (Terminal A)
sudo fail2ban-client status sshd

# Expected output setelah serangan:
# Status for the jail: sshd
# |- Filter
# |   |- Currently failed: X
# |   '- Total failed: X
# '- Actions
#     |- Currently banned: 1
#     '- Banned IP list: 192.168.10.101

# Cek juga fail2ban.log:
sudo grep '192.168.10.101' /var/log/fail2ban.log

```



```

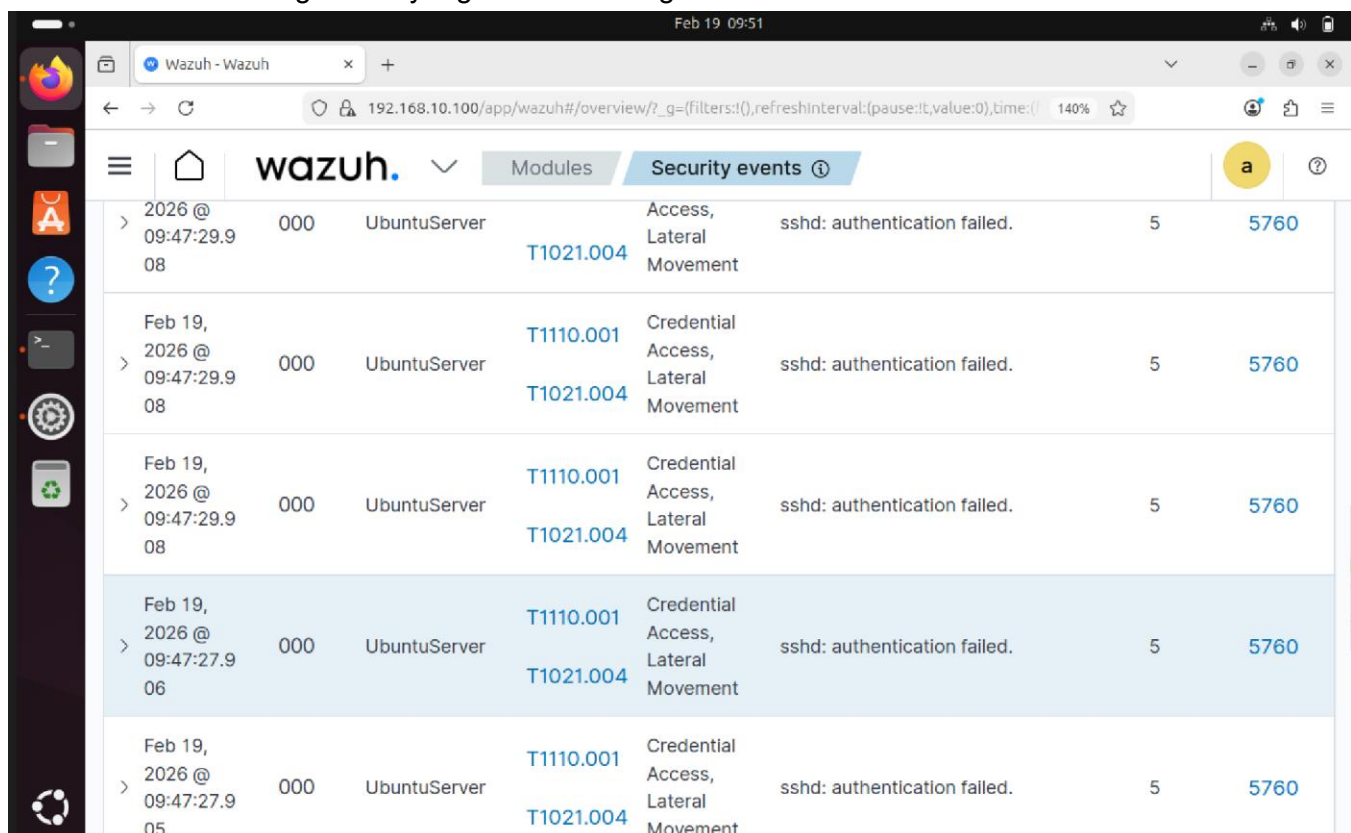
ubuntu@UbuntuServer:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
- Filter
  - Currently failed: 1
  - Total failed: 4
  - Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
- Actions
  - Currently banned: 1
  - Total banned: 1
  - Banned IP list: 192.168.10.101
ubuntu@UbuntuServer:~$ _

```

8.2 Cari Event Fail2ban di Wazuh Dashboard

Wazuh juga memonitor fail2ban.log. Cari event yang menunjukkan Fail2ban melakukan ban terhadap IP attacker.

1. Di Wazuh Dashboard, masuk ke Security Events.
2. Filter dengan keyword: fail2ban atau ban.
3. Cari event dengan rule yang berkaitan dengan Fail2ban action.

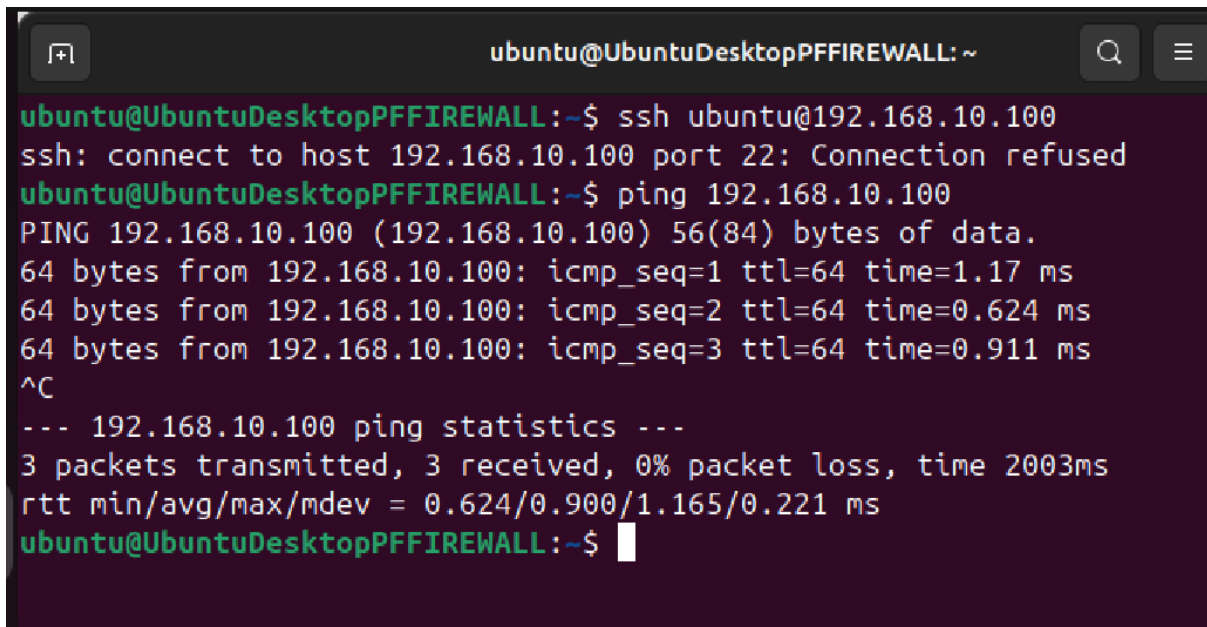


Time	Source	Destination	Rule	Action	Count	IP
Feb 19, 2026 @ 09:47:29.908	192.168.10.101	192.168.10.101	T1110.001 T1021.004	ssh: authentication failed.	5	5760
Feb 19, 2026 @ 09:47:29.908	192.168.10.101	192.168.10.101	T1110.001 T1021.004	ssh: authentication failed.	5	5760
Feb 19, 2026 @ 09:47:29.908	192.168.10.101	192.168.10.101	T1110.001 T1021.004	ssh: authentication failed.	5	5760
Feb 19, 2026 @ 09:47:27.906	192.168.10.101	192.168.10.101	T1110.001 T1021.004	ssh: authentication failed.	5	5760
Feb 19, 2026 @ 09:47:27.905	192.168.10.101	192.168.10.101	T1110.001 T1021.004	ssh: authentication failed.	5	5760

8.3 Verifikasi Koneksi Ditolak dari Desktop

```
# Jalankan di: Ubuntu Desktop (Attacker)
# Coba SSH setelah IP diblokir:
ssh ubuntu@192.168.10.100
# Expected: ssh: connect to host 192.168.10.100 port 22: Connection refused

# Verifikasi network masih jalan (ICMP tidak diblokir):
ping 192.168.10.100 -c 3
# Expected: Reply sukses — hanya SSH yang diblokir
```

A screenshot of a terminal window titled 'ubuntu@UbuntuDesktopPFFIREWALL: ~'. The terminal shows the following commands and output:

```
ubuntu@UbuntuDesktopPFFIREWALL:~$ ssh ubuntu@192.168.10.100
ssh: connect to host 192.168.10.100 port 22: Connection refused
ubuntu@UbuntuDesktopPFFIREWALL:~$ ping 192.168.10.100
PING 192.168.10.100 (192.168.10.100) 56(84) bytes of data.
64 bytes from 192.168.10.100: icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from 192.168.10.100: icmp_seq=2 ttl=64 time=0.624 ms
64 bytes from 192.168.10.100: icmp_seq=3 ttl=64 time=0.911 ms
^C
--- 192.168.10.100 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.624/0.900/1.165/0.221 ms
ubuntu@UbuntuDesktopPFFIREWALL:~$
```

9. Phase 8 — Recovery

Unban IP dan verifikasi sistem kembali normal

9.1 Manual Unban IP Attacker

Dalam skenario nyata, recovery dilakukan ketika admin memverifikasi bahwa IP yang terblokir adalah legitimate. Proses unban bisa dilakukan secara manual atau menunggu bantime (600 detik) habis.

```
# Jalankan di: Ubuntu Server
sudo fail2ban-client set sshd unbanip 192.168.10.101

# Verifikasi unban berhasil:
sudo fail2ban-client status sshd
# Expected: Currently banned: 0, Banned IP list: (kosong)
```

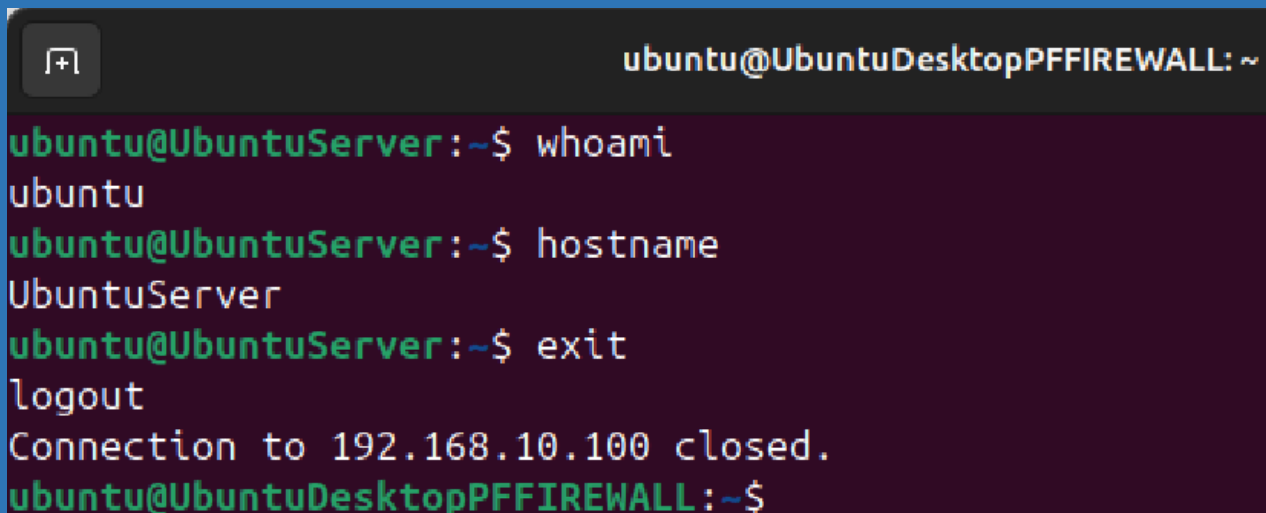
```
ubuntu@UbuntuServer:~$ sudo fail2ban-client set sshd unbanip 192.168.10.101
[sudo] password for ubuntu:
Sorry, try again.
[sudo] password for ubuntu:
0
ubuntu@UbuntuServer:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:     10
|   \- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
|- Actions
|   |- Currently banned: 0
|   |- Total banned:     2
|   \- Banned IP list:
ubuntu@UbuntuServer:~$
```

9.2 Verifikasi Akses SSH Kembali Normal

```
# Jalankan di: Ubuntu Desktop
# Coba SSH dengan password yang benar:
ssh ubuntu@192.168.10.100

# Masukkan password yang benar saat diminta
# Expected: Login berhasil

# Setelah masuk, verifikasi identitas:
whoami
hostname
exit
```

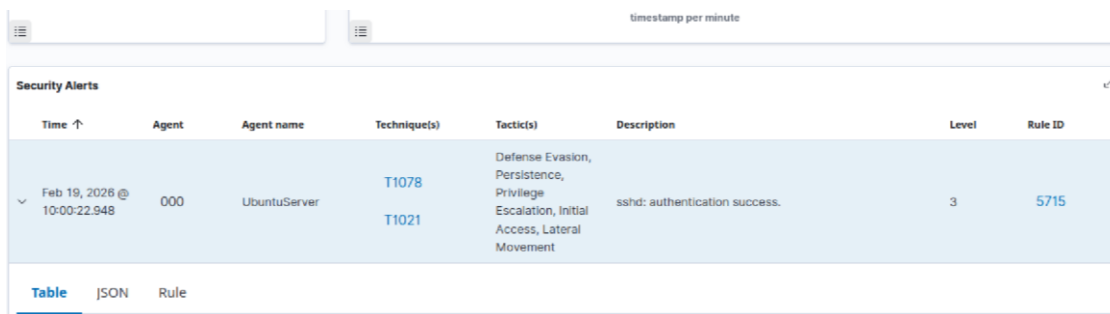


```
ubuntu@UbuntuDesktopPFFIREWALL: ~
ubuntu@UbuntuServer:~$ whoami
ubuntu
ubuntu@UbuntuServer:~$ hostname
UbuntuServer
ubuntu@UbuntuServer:~$ exit
logout
Connection to 192.168.10.100 closed.
ubuntu@UbuntuDesktopPFFIREWALL:~$
```

9.3 Observasi Login Sukses di Wazuh Dashboard

Wazuh juga akan mencatat event login yang berhasil. Cari event ini di dashboard sebagai bukti recovery berhasil.

1. Di Wazuh Dashboard, buka Security Events.
2. Filter dengan keyword: Accepted atau authenticated.
3. Cari event login berhasil dari 192.168.10.101 setelah waktu unban.



The screenshot shows the Wazuh Security Alerts interface. At the top, there's a search bar and a 'timestamp per minute' filter. Below, a table lists security alerts. One alert is highlighted, showing a successful SSH login for agent '000' (UbuntuServer) at 'Feb 19, 2026 @ 10:00:22.948'. The alert details include techniques T1078 and T1021, tactics like Defense Evasion and Privilege Escalation, and a description 'sshd: authentication success.' with a level of 3 and rule ID 5715.

Time ↑	Agent	Agent name	Technique(s)	Tactics	Description	Level	Rule ID
Feb 19, 2026 @ 10:00:22.948	000	UbuntuServer	T1078 T1021	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Lateral Movement	sshd: authentication success.	3	5715

10. Phase 9 — Final System Check

Memastikan semua layanan kembali normal

10.1 Verifikasi Semua Service Masih Aktif

```
# Jalankan di: Ubuntu Server
# Cek semua layanan:
sudo systemctl status ssh
sudo systemctl status fail2ban
sudo systemctl status wazuh-manager
sudo systemctl status wazuh-dashboard
sudo systemctl status wazuh-indexer

# Cek UFW masih aktif:
sudo ufw status verbose
```

Yang harusnya terlihat:

- ✓ ssh: active (running)
- ✓ fail2ban: active (running)
- ✓ wazuh-manager: active (running)
- ✓ wazuh-dashboard: active (running)
- ✓ wazuh-indexer: active (running)
- ✓ UFW: Status: active

11. Incident Report Card & Kesimpulan

Dokumentasi akhir praktikum

Incident Report Card

Incident ID	IR-2026-002
Tanggal	19/02/2026
Severity	Medium
Jenis Serangan	SSH Brute Force Attack
Attacker IP	192.168.10.101 (Ubuntu Desktop)
Target IP	192.168.10.100 (Ubuntu Server)
Tool Digunakan	Hydra (wordlist: rockyou.txt)
Terdeteksi Oleh	Wazuh SIEM + auth.log + fail2ban.log
Response Otomatis	Fail2ban auto-ban setelah 3 failed attempts
Status Akhir	Contained — Tidak ada akses berhasil
Ditangani Oleh	Glenvio Regalito Rahardjo

Kesimpulan

Praktikum Modul 3 ini berhasil membuktikan bahwa integrasi Wazuh SIEM memperkuat sistem Defense in Depth yang sudah dibangun pada modul sebelumnya. Dengan menambahkan SIEM sebagai layer keempat, sistem tidak hanya mampu memblokir serangan secara otomatis, tetapi juga memberikan visibilitas terpusat yang dibutuhkan oleh SOC analyst.

Melalui Wazuh Dashboard, seluruh timeline insiden dari awal percobaan brute force, deteksi pola berbahaya, tindakan Fail2ban, hingga recovery dapat dipantau dan didokumentasikan dari satu antarmuka. Ini merepresentasikan mini SOC environment berbasis open-source yang relevan dengan praktik industri keamanan siber nyata.

Key Takeaway dari Trilogy Defense in Depth Lab Project

Modul 1 (Defense in Depth): Membangun fondasi keamanan berlapis dengan UFW, SSH Hardening, dan Fail2ban

Modul 2 (Brute Force IR): Membuktikan sistem mampu mendeteksi dan merespons serangan secara otomatis menggunakan framework NIST SP 800-61

Modul 3 (Wazuh SIEM): Menambahkan visibilitas terpusat melalui SIEM — mengintegrasikan semua layer menjadi satu dashboard SOC

Kombinasi ketiga modul ini mencerminkan implementasi keamanan server yang profesional dan defensif.

Ethical & Legal Reminder

Seluruh aktivitas dalam modul ini dilakukan pada sistem milik sendiri dalam lab terisolasi. Brute force attack, port scanning, dan eksploitasi terhadap sistem tanpa izin adalah tindakan ilegal yang dapat dikenakan sanksi hukum. Gunakan pengetahuan ini hanya untuk defensive security dan ethical hacking yang terotorisasi.